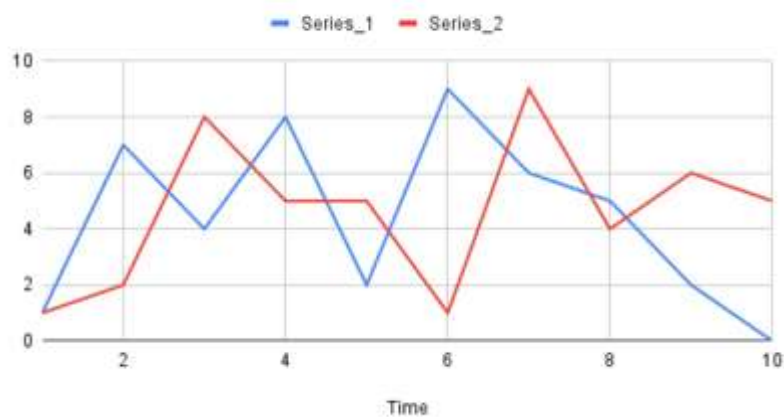


How DTW Algorithm work?

Let us take two-time series signals A and B:

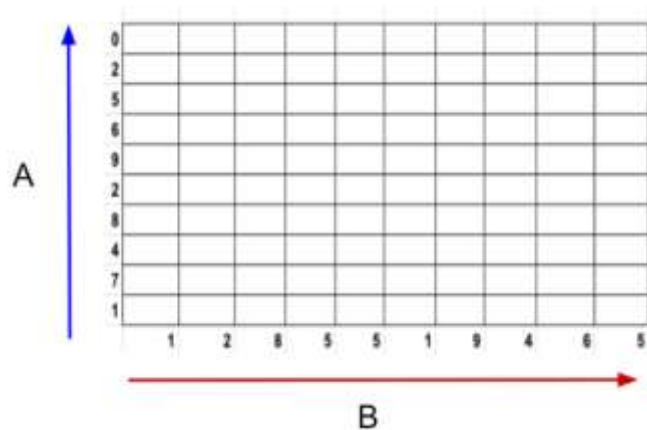
Time	Series_1	Series_2
1	1	1
2	7	2
3	4	8
4	8	5
5	2	5
6	9	1
7	6	9
8	5	4
9	2	6
10	0	5

Series_1 and Series_2



Step 1: Empty Cost Matrix Creation

Create an empty cost matrix M with x and y labels as amplitudes of the two series to be compared.



Step 2: Cost Calculation

Fill the cost matrix using the formula mentioned below starting from the left and bottom corners.

$$M(i, j) = |A(i) - B(j)| + \min (M(i-1, j-1), M(i, j-1), M(i-1, j))$$

Where,

M is the matrix

i is the iterator for series A

j is the iterator for series B

So in the below table, we have computed the cost of some values.

0	36	29								
2	35	27								
5	34	27								
6	30	24								
9	25	20								
2	17	13								
8	16	13	6	6						
4	9	7	6	3						
7	6	5	2	4	6	12	14	17	18	20
1	0	1	8	12	16	16	24	27	32	36
	1	2	8	5	5	1	9	4	6	5

Let us take a few examples from the above table (6, 9, and 12) to illustrate the calculation as highlighted in the below table

For 6,

0	36	29								
2	35	27								
5	34	27								
6	30	24								
9	25	20								
2	17	13								
8	16	13	6	6						
4	9	7	6	3						
7	6	5	2	4	6	12	14	17	18	20
1	0	1	8	12	16	16	24	27	32	36
	1	2	8	5	5	1	9	4	6	5

$$|8 - 5| + \min(6, 6, 3) = 6$$

Similarly for 9,

$$|4 - 1| + \min(6) = 9$$

And the same for 12,

$$|1 - 5| + \min(8) = 12$$

The full table will look like this:

0	36	29	32	22	22	16	24	16	18	17
2	35	27	24	17	17	15	21	12	14	12
5	34	27	18	14	14	18	14	10	10	9
6	30	24	15	14	14	18	10	9	9	10
9	25	20	13	13	13	15	7	12	13	17
2	17	13	12	9	9	7	14	10	14	17
8	16	13	6	6	6	11	8	12	14	17
4	9	7	6	3	4	7	12	12	14	15
7	6	5	2	4	6	12	14	17	18	20
1	0	1	8	12	16	16	24	27	32	36
	1	2	8	5	5	1	9	4	6	5

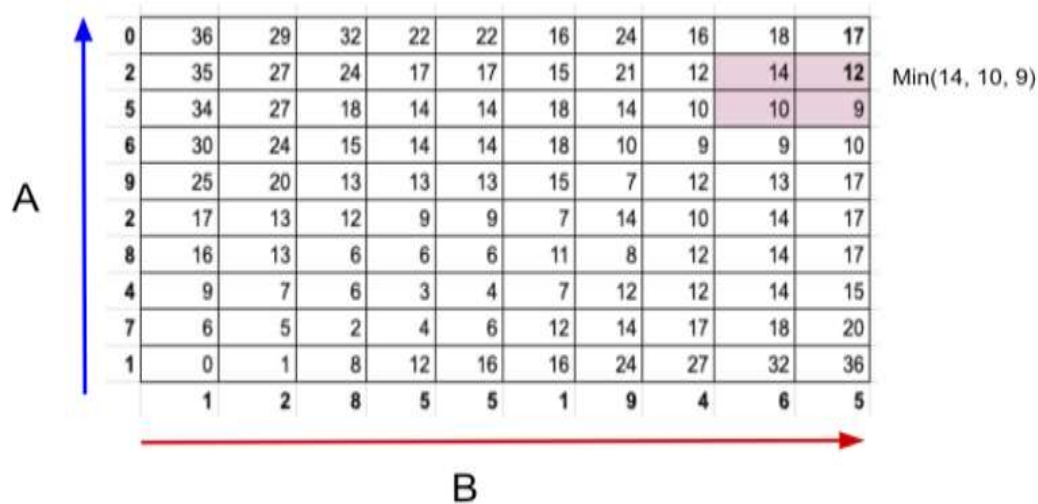
Step 3: Warping Path Identification

Identify the warping path starting from the top right corner of the matrix and traversing to the bottom left. The traversal path is identified based on the neighbour with a minimum value.

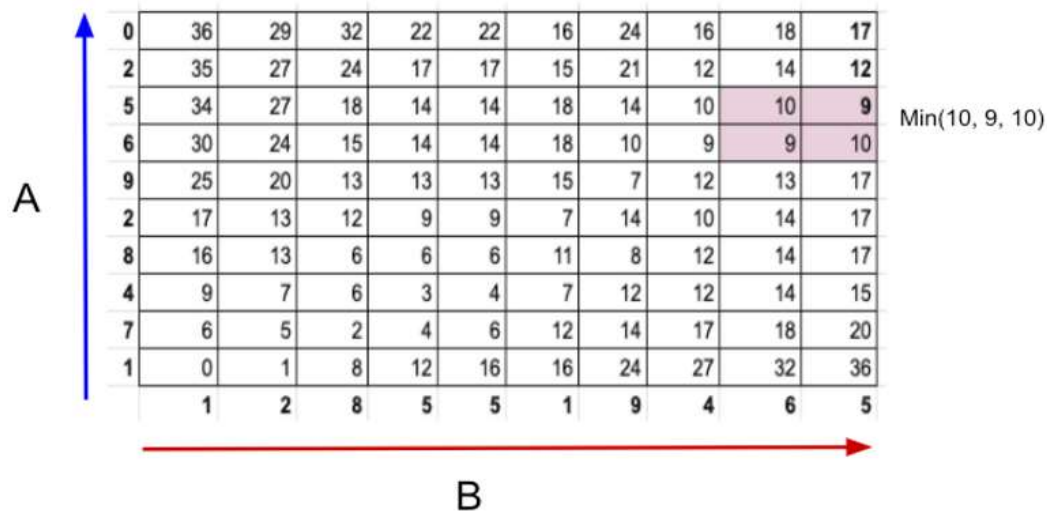
In our example, it starts with 17 and looks for the minimum value i.e. 12 among its neighbours 18, 14, and 12.

0	36	29	32	22	22	16	24	16	18	17
2	35	27	24	17	17	15	21	12	14	12
5	34	27	18	14	14	18	14	10	10	9
6	30	24	15	14	14	18	10	9	9	10
9	25	20	13	13	13	15	7	12	13	17
2	17	13	12	9	9	7	14	10	14	17
8	16	13	6	6	6	11	8	12	14	17
4	9	7	6	3	4	7	12	12	14	15
7	6	5	2	4	6	12	14	17	18	20
1	0	1	8	12	16	16	24	27	32	36
	1	2	8	5	5	1	9	4	6	5

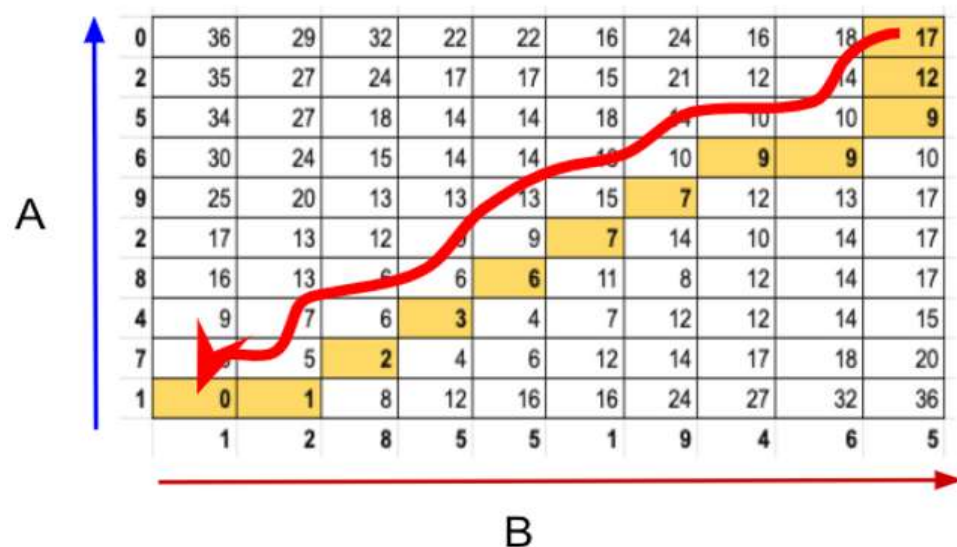
Min(18, 14, 12)



The next number in the warping traversal path is 9. This process continues till we reach the bottom of the left axis of the table.



The final path will look like this:



Let this warping path series be called as d.

d = [17, 12, 9, 9, 9, 7, 7, 6, 3, 2, 1, 0]

Step 4: Final Distance Calculation

Time normalized distance, D

$$D = \frac{\sum_{i=1}^k d(i)}{\sum_{i=1}^k k}$$

Where k is the length of the series d

k = 12 in our case.

D = (17 + 12 + 9 + 9 + 9 + 7 + 7 + 6 + 3 + 2 + 1 + 0) / 12

D = 6.8333